

QVI CUSTOMER CHIP PURCHASING BEHAVIOUR ANALYSIS REPORT

Analyst: Joy Ikwue

Date: June, 2026

Programme: QUANTIUM (Forage)

Dataset: Transaction Data (264,836 rows) & Customers Data
(72,638 rows)

Tools: Excel

EXECUTIVE SUMMARY

\$1.9M	504.7K	\$3.8	21	264.83K
Total Revenue	Quantity Units Sold	Average Price	Customer Seg.	Transactions

This report presents a data-driven analysis of QVI customer chip purchasing behaviour for the period 2018–2019. The dataset was cleaned to remove outliers, correct inconsistent formats and remove duplicate values to ensure data integrity. Five key metrics were analysed across 21 customer segments using pivot tables in Excel. The analysis revealed that Older Families and Young Families are the most valuable and frequently shopping segments, while Mainstream Young Singles/Couples demonstrated the highest willingness to pay per unit. These findings provide actionable insights to support Julia, Category Manager, in making informed decisions around promotional strategy, range planning and chip category growth for the upcoming category review.

PROBLEM STATEMENT

The QVI chips category is approaching a scheduled category review and the Category Manager, Julia, requires a clear understanding of customer purchasing behaviour to inform strategic decisions. While transactional data exists across multiple customer segments, there is currently no structured analysis identifying which customer segments drive the most value, how frequently different segments purchase, and what factors influence their spending behaviour. Without this understanding, decisions around range planning, promotional strategy and category growth risk being made without sufficient data backing. This analysis therefore seeks to examine chip purchasing behaviour across 21 customer segments defined by lifestage and premium tier, in order to provide Julia with actionable and commercially relevant recommendations for the upcoming category review.

DATA OVERVIEW

Two datasets were used in this analysis. The first dataset contained transactional data recording 264,836 rows of chip purchases made between July 2018 and June 2019 across multiple stores. Each transaction recorded the date, store number, customer loyalty card

number, transaction ID, product number, product name, quantity purchased and total sales value.

The second dataset contained customer behaviour data covering 72,637 unique customers, each classified by two attributes, lifestage and premium tier. Lifestage identifies where a customer is in their life journey, capturing seven categories including Young Singles/Couples, Young Families, Midage Singles/Couples, Older Families, Older Singles/Couples, Retirees and New Families. Premium tier differentiates customers by their price point preferences across three categories which are Budget, Mainstream and Premium.

Both datasets were joined using the customer loyalty card number as the primary key, resulting in a single merged dataset ready for analysis. Prior to merging, the transaction dataset was cleaned to remove 2 bulk purchase transactions belonging to a commercial buyer (Customer 226000) and 1 duplicate transaction, leaving a final clean dataset of 264,833 transactions. No missing values or formatting issues were identified in the customer behaviour dataset.

Data Dictionary

Transaction Data

Variable	Type	Description	Used in Analysis
Date	Date	The date the transaction was made	Yes
Store Number	INT	Unique identifier for each store	No
Loyalty Card Number	INT	Unique customer loyalty card number	Yes
Transaction ID	INT	Unique identifier for each transaction	Yes
Product Number	INT	Unique identifier for each product	No
Product Name	Varchar	Full name of the chip product purchased	No
Product Quantity	INT	Number of units purchased per transaction	Yes
Unit Price	INT	Derived variable — Unit price per unit (TOT_SALES ÷ PROD_QTY)	Yes
Total Sales	INT	Total sales value of the transaction in dollars	Yes

Customer Data

Variable	Type	Description	Used in Analysis
LYLTY_CARD_NBR	INT	Unique customer loyalty card number	Yes
LIFESTAGE	Varchar	Customer lifestage segment — 7 categories identifying family and life stage status	Yes
PREMIUM_CUSTOMER	Text	Customer premium tier — Budget, Mainstream or Premium based on price point preferences	Yes

Data Quality Issues & Decisions

Issue	Volume	Decision
Outlier detected - Commercial bulk buyer (Customer 226000) purchasing 200 units per transaction	2 rows	Removed from dataset to prevent skewing of analysis results
Duplicate transaction record	1 row	Removed to avoid double counting in analysis
Inconsistent data format - DATE column stored as serial numbers	All rows	Corrected by reformatting column to DD/MM/YYYY date format

METHODOLOGY

A six-stage pipeline using Excel was adopted to ensures reproducibility and separation of raw from analytical data.

Stage	Tool	Output	Rationale
Data Ingestion	Excel	Two datasets loaded - transaction data (264,836 rows) and customer behaviour data (72,637 rows)	Established the foundation for all subsequent analysis
Data Exploration	Excel	High level summaries, row counts, column types and unique value checks performed across both datasets	Identified data quality issues prior to cleaning
Data Cleaning & Transformation	Excel	Removal of 2 outlier rows, 1 duplicate row	Ensured data integrity and

		and correction of DATE format. Derived variables UNIT_PRICE, PACK_SIZE.	enriched the dataset for deeper analysis
Data Merging	Excel	Transaction and customer datasets merged using XLOOKUP on LYLTY_CARD_NBR. LIFESTAGE and PREMIUM_CUSTOMER columns added to transaction dataset	Created a single unified dataset enabling customer segment analysis
Analysis	Excel – Pivot table	Five pivot tables created measuring Total Sales, Number of Customers, Transactions per Customer, Units per Transaction and Average Price per Unit across 21 customer segments	Addressed the key business questions around customer purchasing behaviour.
Visualization & Reporting	Excel	Summary presentation sheet created for Julia with tables, highlights and strategic recommendations	Translated analytical findings into commercially actionable insights

KEY FINDINGS

1. Mainstream is the Dominant Revenue Driver by Price Tier

Mainstream customers generated the highest total revenue at \$750,744, accounting for 38.8% of total category sales. This was driven largely by the sheer volume of Mainstream customers at 29,245, the largest price tier by customer count. This suggests that category revenue is primarily volume driven rather than premium spend driven.

Implication: *Mainstream customers are the backbone of the chips category. Any promotional or ranging strategy must prioritise this tier to protect the category's revenue base.*

2. Older and Young Families are the Highest Value Individual Customers

Despite not being the largest segments by customer count, Older Families and Young Families recorded the highest sales per customer at \$36.04 and \$34.45 respectively. Both segments also shop most frequently at 4.95 and 4.75 transactions per year and buy the most units per visit at 1.95 and 1.94 units per transaction.

Implication: *Families represent the most engaged and loyal shoppers in the category. Strategies targeting multipack deals, larger pack sizes and family promotions will resonate strongly with this segment and drive both frequency and basket size.*

3. Retirees and Older Singles/Couples Drive Volume

Retirees and Older Singles/Couples are the largest segments by customer count at 14,805 and 14,609 respectively, collectively contributing \$768,892 in total sales nearly 40% of total category revenue. Their purchasing behaviour is consistent and stable across all metrics.

Implication: *These segments are critical to category health. Retention strategies focused on range availability, shelf presence and loyalty rewards are essential to maintaining this revenue base.*

4. Mainstream Young Singles/Couples Show Highest Price Willingness

Mainstream Young Singles/Couples paid the highest average price per unit at \$4.07 above the overall category average of \$3.82. Despite having the lowest transaction frequency at 2.58 visits per year, this segment has the third largest customer base at 8,088 Mainstream customers alone.

Implication: *Young Singles/Couples are brand driven rather than price driven. There is a significant opportunity to increase their purchase frequency through targeted marketing, new flavour launches and premium brand placements at impulse purchase locations.*

5. New Families is the Least Engaged Segment

New Families recorded the lowest total sales at \$50,433, the smallest customer base at 2,549 and one of the lowest transaction frequencies at 2.70 visits per year. This pattern was consistent across all three premium tiers.

Implication: *New Families may be underserved by the current chip range or may simply be prioritising other food categories at this life stage. Further investigation is recommended to determine whether this represents an untapped opportunity or a natural category limitation.*

RECOMMENDATION

1. Grow the Families Segment

Introduce multipack deals, larger pack sizes and value for money promotions targeting Older and Young Families. Given their high purchase frequency and price conscious

behaviour, family bundle promotions placed in high traffic store locations will effectively drive both basket size and visit frequency.

Priority: High.

2. Unlock Young Singles/Couples Purchasing Frequency

Target Mainstream Young Singles/Couples with new flavour launches, premium brand placements and impulse purchase displays. Their high willingness to pay and large customer base makes increasing their low visit frequency of 2.58 times per year the single biggest growth opportunity in the category.

Priority: High

3. Maintain Retirees and Older Singles/Couples

Ensure consistent range availability, strong shelf presence and loyalty rewards to protect this segment. Collectively contributing nearly 40% of total revenue, retention of these customers is critical to category health.

Priority: Medium

4. Investigate New Families

Conduct further research to determine whether the low engagement from New Families reflects an underserved opportunity or a natural category limitation before committing any significant investment to this segment.

Priority: Low

CONCLUSION

This analysis examined chip purchasing behaviour across 21 customer segments for the period 2018–2019, providing a data-driven foundation for Julia's upcoming category review. The findings reveal that while Mainstream customers drive category revenue through volume, Older and Young Families deliver the highest individual customer value through frequent and high volume purchasing. Mainstream Young Singles/Couples present the most significant growth opportunity given their large customer base and demonstrated willingness to pay. Retirees and Older Singles/Couples remain the backbone of the category and must be retained through consistent ranging and loyalty initiatives. By prioritising the four strategic recommendations outlined in this report, Julia and the QVI team are well positioned to drive sustainable chip category growth, improve customer engagement across key segments and make informed decisions heading into the next category review period.